Stick-together calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a common final velocity using total momentum.

SUCCESS CRITERIA - I can...

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “inelastic” and “initial” to explain your model or result. Sentence starters are optional.

inelastic: a collision that transforms some kinetic energy to other forms

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

01 Bring back an earlier idea

Fragments of masses 3 and 5 kg separate from rest. The 3 kg piece moves at +10 m/s.
Find the other velocity and total kinetic energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. v_f is the shared final velocity.

Worked reasoning: A 2 kg cart at +3 m/s sticks to a 1 kg cart at rest. Total momentum 6 kg·m/s divided by total mass 3 kg gives final velocity +2 m/s.

Your turn: Mark the conserved quantity on a before-and-after sketch.
Explain why masses are added in the final denominator.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two carts stick.
Identify which masses belong in the final momentum expression and why.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Solve and justify

A 2 kg cart at +5 m/s sticks to a 3 kg cart at -1 m/s.
Find final velocity and kinetic energy transformed.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student averages the two starting velocities without considering mass. When would that happen to work?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

At fixed mass and speed, circle radius becomes 4 times as large.
Compare acceleration and net force using equations.

Word help: net force = the vector sum of all forces on one object

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

A .20 kg projectile at 30 m/s sticks to a 1.80 kg cart at rest.
Find common speed and fraction of initial kinetic energy remaining.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 1 kg cart at +6 m/s sticks to a 2 kg cart at -3 m/s.
Find final velocity. If the second velocity were -1 m/s instead, find the new result.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3